Enrico **Veraldi**



Profile

I am a PhD student in Astrophysics and Cosmology at SISSA (Trieste)and a master's student in High Performance Computing (MHPC) at ICTP (Trieste).
My main research interests regard the characterization of the physical properties of the interstellar medium in high-redshift galaxies and the creation of associated models.

Contact details

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Personal information

- Born on 15/09/199, Bologna (Italy)
- English (B2), Italian (native)

Tech Stack

- Python, Fortran, C/C++, CUDA, Bash, R
- Numpyh, Pandas, Matplotlib, Scikit-learn, PyTorch
- LaTeX, Docker, Anaconda
- Hugging Face, Ollama
- CLOUDY, NRAO CASA, ds9
- MPI, OpenMP, OpenACC
- Linux Systems, Slurm, Base System Adm.

Present Positions

PhD. Student
SISSA, Trieste (TS). **2024.10–pres.**
◇ PhD in Astrophysics and Cosmology (APC sector)

Master's Student
MHPC IFTP-SISSA, Trieste (TS). **2025.09–pres.**
◇ 2nd level master's in high performance computing

Education

Master in Astrophysics and Cosmology
University of Bologna (UNIBO). **2021–2023**
◇ Thesis title: *The ISM properties of ALPINE galaxies as probed by ALMA and JWST: a comprehensive line emission modelling.*
◇ Supervisors: Prof. Francesca Pozzi, Dr. Livia Vallini, Dr. Federico Esposito.

Bachelor in Physics
University of Bologna (UNIBO). **2018–2021**
◇ Thesis title: *Study of a spiral galaxy in the Local Universe.*
◇ Supervisors: Prof. Francesca Pozzi, Dr. Viviana Casasola.

High school graduation
Scientific High School Niccolò Copernico (Bologna). **2018–2021**
◇ Scientific Applied Sciences

First-author Publications

The ALPINE-ALMA [CII] Survey: Modelling ALMA and JWST lines to constrain the interstellar medium of $z \sim 5$ galaxies **2025.01**
◇ Veraldi et al. 2025, DOI: 10.1051/0004-6361/202451111

Co-author Publications

The ALPINE-CRISTAL-JWST Survey: The Fast Metal Enrichment of Massive Galaxies at $z \sim 5$ **2026.06**
◇ Faisst et al. 2026, DOI: 10.3847/1538-4357/ae6645

The ALPINE-CRISTAL-JWST Survey: Chemical Abundance Comparison between the ISM and CGM of Main-sequence Galaxies at $z = 4 - 6$ **2026.04**
◇ Wang et al. 2026, DOI: 10.3847/1538-4357/ae4b38

The ALPINE-CRISTAL-JWST Survey: JWST/IFU Optical Observations for 18 Main-sequence Galaxies at $z = 4 - 6$ **2026.01**
◇ Faisst et al. 2026, DOI: 10.3847/1538-4365/ae0928

The ALPINE-CRISTAL-JWST survey: revealing less massive black holes in high-redshift galaxies **2025.11**
◇ Ren et al. 2025, DOI: 10.1093/mnras/staf1709

The ALPINE-CRISTAL-JWST survey: Spatially resolved star formation relations at $z \sim 5$ **2025.10**
◇ Accard et al. 2025, DOI: 10.1051/0004-6361/202556140

Spatially resolved [CII]-gas conversion factor in early galaxies **2025.08**
◇ Vallini et al. 2025, DOI: 10.1051/0004-6361/202555179

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Referees

@ Prof. Andrea Lapi

@ Dr. Tommaso Ronconi

@ Dr. Livia Vallini

@ Prof. Francesca Pozzi

Posters/Talks/Seminars

Seminar "A Panchromatic View of the Early ISM: Modelling UV, Optical, and FIR Lines at z 5 connecting ALMA and JWST" **2026.05**
◇ Seminar: "Astrophysics talk", INAF-OAS Bologna (Italy) (website)

Talk "The ALPINE-ALMA [CII] Survey: Modelling ALMA and JWST lines to characterise the interstellar medium of high-redshift galaxies" **2025.12**
◇ Conference: "The Baryon Cycle from Reionization to the Cosmic Noon", Puerto Varas (Cile) (website)

Poster-Flash talk "The ALPINE-ALMA [CII] Survey: Modelling ALMA and JWST lines to characterise the interstellar medium of high-redshift galaxies" **2025.06**
◇ Conference: "Sixth Workshop on Millimetre Astronomy in Italy", Bologna (ARC)

Poster-Flash talk "The ISM properties of ALPINE galaxies as probed by ALMA and JWST" **2024.09**
◇ Conference: "Views on the multi-phase interstellar medium in galaxies", Bologna (poster)

Schools/Workshops

AIPHY2: Bayesian Statistics. Doctoral school on Bayesian Statistics **2026.10**
◇ Gran Sasso Science Institute (GSSI), l'Aquila (Italy) (website)

Summer School "Statistics for Astronomers" **2026.06**
◇ Center for Astrostatistics at The Pennsylvania State University, PA 16802 (USA) (website)

Workshop "Probing Galaxy Evolution at Cosmic Noon with Lensed DSFGs: A Multi-Wavelength Approach from Far-infrared to Radio" **2026.04**
◇ IFPU, Trieste (Italy) (website)

Scientific Collaborations

Member GoTha Group, SISSA (website) **2025-pres.**

Member ALPINE-CRISTAL-JWST Collaboration **2024-pres.**

Member of Hybrid PRIN project AstroMIND (website) **2026-pres.**